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Cloud Computing Assignment - 2

Ques. 1. Explain the architecture of the helm chart, including its main components?

=) A helm chart is a packaged used in kubernetes to define, install and manage applications.

Main components

1) Chart. yaml
- Contains metadata (name, version, description)
- Defines dependencies.

2) Values. yaml
- Default configuration values
- Can be overidden during deployment

3) templates/
- Contains kubernetes manifest templates (YAML files)
- Uses go templating for dynamic values.

4) Charts/
- stores dependent charts

5) · helmignore
- specifies files to exclude from packaging

6) Notes. txt
- Provides post-install instructions to users.

Ques. 2. Discuss the advantages and limitations of serverless (FaaS)?

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⇒ Advantages:
- No server management - handled by providers like Amazon Web Services or google cloud.
- Auto-scaling: Scales based on demand.
- Cost-efficient: Pay only for execution time.
- Faster development: focus on business logic
- High availability: Built-in fault tolerance

Limitations:
- Cold start latency
- Execution time limits
- Vendor lock-in
- Limited control over infrastructure
- Debugging complexity.

Ques. 3. Explain how Google BigQuery handles large-scale analytics efficiently?

⇒ Google BigQuery is designed for massive data analytics.

Key features:
1) Columnar Storage
- stores data column wise → faster queries

2) Distributed processing
- Uses parellel execution model for fast aggregation.

3) Dremel engine
- Tree based execution model for fast aggregation.

4) Serverless architecture

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- No infrastructure management required

5) Seperation of storage and compute
- Independent scaling

6) Query optimization
- Automatic optimization of execution plans

7) Built-in ML (Big Query ML)
- Run ML directly on data without exporting

Ques.4. Compare:
- Pre-built ML APIs
- Auto ML
- Custom ML models.

| ⇒ Feature | Pre-built ML APIs | Auto ML | Custom ML models |
|---|---|---|---|
| Definition | Ready made APIs | Semi-automated ML | Fully custom built |
| Skill req. | Low | Medium | high |
| Flexibility | Low | Medium | high |
| Use case | Generic tasks | Domain-specific tuning | Complex problems |
| Examples | Vision API, NLP API | Auto ML vision | Tensor flow models |
| Dev. time | Very fast | Moderate | long |

Ques.5. Explain the complete lifecycle of a serverless function, including trigger, execution and scaling?

⇒ 1) Trigger phase
- Event occurs (HTTP request, file upload, database change)
- Function is invoked

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2) Initialization (Cold start)
- Runtime environment is created
- Dependencies are loaded

3) Execution
- Func. code runs
- Processes input and generates output

4) Response
- Output returned to caller or stored

5) Scaling
- Automatically scales based on incoming requests
- Multiple instances run in parellel.

6) Termination
- Idle instances are shut down to save cost.

Ques. 6. Discuss the role of helm in DevOps and CI/CD pipelines?
=) Helm plays a crucial role in automation.

Key roles:
1) Application packaging
- Standadizes deployment artifacts

2) Environment management
- Different configs via values.yaml (dev, staging, prod)

3) Version control
- Manage app. versions and rollbacks.

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4) CI/CD integration
- Used with Tools like Jenkins or Github Actions

5) Automated deployments
- Enables repeatable and consistent deployments

6) Rollback capability
- Easy recovery from failures

Ques. 7. Explain the working of edge inference with a step-by-step process?

⇒ Edge interference means running AI models locally on devices instead of cloud.

Step-by-step process:
1) Model training
- Model is trained in cloud (eg. using GPUs)

2) Model optimization
- Compression, quantization for smaller size.

3) Deployment to edge device
- Devices like smartphone, IOT devices

4) Data collection
- Real-time data from sensors/cameras

5) Local inference
- Model processes data on device

Date .....

6) Decision making
- Immediate response (eg. object detection)

7) Optional cloud sync
- Send results or updates to cloud

Advantages
- Low latency
- Improved privacy
- Reduced bandwidth usage.